

have been raised about their safety and suitability, especially for products destined for human contact.

Researchers generated excitement in 2006, when the destructive nature of certain nanoparticles was found to destroy the cell membranes of cancer cells. In the culture used in the early experiments, healthy cells were affected, but less so than cancer cells. The researchers, in their excitement, suggested that these nanoparticles be used to enclose chemotherapy agents, thus target the cancer cells more directly. The early study, however, only surveyed the nanoparticles in isolation, without the toxic chemicals enclosed in them. However a recent study, presented in the nano journal `Small`, showed that cell cultures (colonies of a particular type of cell growing in a dish) are unaltered when exposed to fullerenes. The same cells do not react when exposed to Gallic acid, an astringent component of tannins found in almost all plants. When present in the cell culture at the same time, however, fullerenes and gallic acid interact to form structures which bind to the cell's surface causing cell death. Hence, it stands to reason that when nanoparticles interact with other particles, unforeseen events might follow. Combining these particles which do not behave as we expect with the already dismal failure of chemotherapy seems like a recipe for disaster.

Regulation

Because of the ongoing argument on the implications of nanotechnology, there is significant debate concerning whether nanotechnology or nanotechnology-based products merit special government regulation.

The nanotechnology label is used on an increasing number of commercially available products – from socks and trousers to tennis racquets and cleaning cloths. The emergence of such nanotechnologies, and their accompanying industries, has triggered calls for increased regulatory arrangements. However, these calls have presently not led to any comprehensive regulation in order to oversee research and the commercial application of nanotechnologies, or any comprehensive labeling for products that contain nanoparticles or are derived from nano-processes.

Although some non-nanotechnology specific regulatory agencies currently cover some products and processes (to varying degrees) – by “bolting on” nanotechnology to existing regulations – there are clear gaps in these regimes. This enables some nanotechnology applications to figuratively “slip through the cracks” without being covered by any regulations.